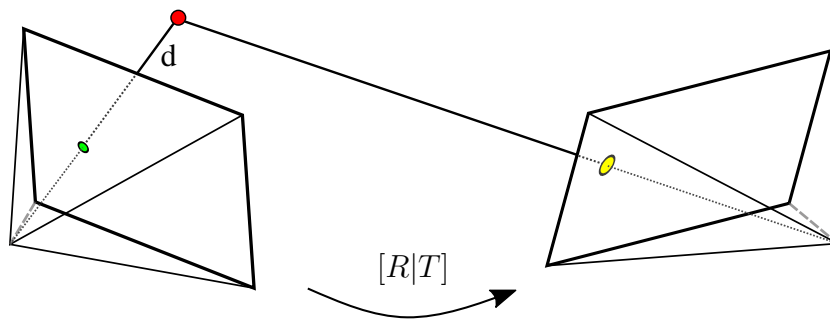


EX2: Reprojector

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Deadline: 31.05.2023 11:00 CET



Problem description

In this task, your objective is to reproject a point from one image onto another (yellow). You will be provided with the location of a point in the first image (green), as well as the depth (d), rotation (R), translation (T), and camera calibration for two images.

The program will take as input

- camera model for the first camera (pinhole/fov)
- camera model for the next camera (pinhole/fov)
- rotation and translation represented in a single 3x4 matrix $[R|T]$ (a transformation from first camera's coordinate system to the second camera's coordinate system)
- the rest of the input are the points on the first image with corresponding depths

pinhole camera is given to you as:

```
pinhole <width> <height> <focal x> <focal y> <center x> <center y>
```

and fov as:

```
fov <width> <height> <focal x> <focal y> <center x> <center y> <w>
```

Your task is to output, in a given order, the corresponding point locations on the other image. If a point lies outside of the image, the output should be "OB".

FOV model

A general image formation model for radially distorted cameras is generic projection followed by a non-linear transformation of the radius for each image point. The distorted coordinates of a generically projected point $\tilde{\mathbf{X}}$ are given by

$$\pi_d(\tilde{\mathbf{X}}) = g\left(\|\pi(\tilde{\mathbf{X}})\|\right) \cdot \pi(\tilde{\mathbf{X}}) \in \mathbb{R}^2.$$

$g : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is the function that radially distorts the coordinates of $\pi(\tilde{\mathbf{X}})$. It is typically approximated by some parametric function. The pixel coordinates of the distorted camera are

$$\begin{pmatrix} u_d \\ v_d \\ 1 \end{pmatrix} = K \begin{pmatrix} \pi_d(\tilde{\mathbf{X}}) \\ 1 \end{pmatrix} = K \begin{pmatrix} g\left(\|\pi(\tilde{\mathbf{X}})\|\right) \cdot \pi(\tilde{\mathbf{X}}) \\ 1 \end{pmatrix}.$$

The so-called FOV model, first suggested by Devernay and Faugeras in 2001, and used e.g. in the open source implementation of PTAM (Parallel Tracking and Mapping), is given by

$$g_{\text{ATAN}}(r) = \frac{1}{\omega r} \arctan\left(2r \tan\left(\frac{\omega}{2}\right)\right).$$